

ALEXANDER MAY

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Experience

Dell Technologies

Hopkinton, MA

Software Engineer

July 2024 - August 2026

- Built LLM-powered diagnostic tools that compute health and update-plan risk for customers' storage assets at query time to **answer natural-language questions in under 7 seconds**
- Extended and optimized Agent2Agent (A2A) routing for an AI agent (LlamaIndex ReAct, Claude on AWS Bedrock) to **reliably execute update workflows on production storage arrays**
- Designed a timestamp tagging protocol and prompt contract that keep timestamps machine-parseable through LLM output to unify localization across A2A stream events and **eliminate client-side timezone bugs**
- Architected an event-driven AMQP pipeline ingesting and enriching **1M+ messages daily** to feed update recommendations into an AI-powered storage platform **in place of manual data loading**
- Engineered Golang controllers that detect and gracefully recover from failed node lifecycle operations on an on-prem Kubernetes data protection appliance to **prevent a disruptive 8-hour cluster redeploy**
- Triaged CVEs, ran STRIDE threat models, and managed quarterly security assessments **across two products as Security Champion**
- Published an AI threat-modeling skill to the organization's shared engineering platform that produces CVSS-scored assessments via a security-scanning Model Context Protocol (MCP) server, **cutting a multi-day process to minutes**
- Stood up the **first production-fidelity penetration testing environment** for an agentic storage platform by replicating **50+ microservices** and authored the **runbook used in subsequent cycles**
- Mentored a team of summer interns** on a chaos engineering project, guiding their selection of Chaos Mesh for Kubernetes-compatible fault injection, reviewing fault design feasibility, and troubleshooting cluster issues

North Carolina State University

Raleigh, NC

Teaching Assistant, E115: Introduction to Computing Environments

August 2022 - May 2024

- Taught Unix/Linux fundamentals (shell navigation, file systems, permissions, remote access) to **~50 students per semester across 4 semesters (200+ total)**
- Ran office hours and lab sessions debugging student code, graded assignments, and worked with course staff on delivery each term

Projects

Chalk Talk, Senior Design, NC State Laboratory for Analytic Sciences

January 2024 - May 2024

- Built a full-stack research web app (**React, FastAPI, PostgreSQL, Docker**) turning highlighted transcript phrases into **Stable Diffusion**-generated conceptual peg imagery in under 3 seconds, with WebVTT parsing, **wavesurfer.js** waveform sync, and persistent media storage

Networked Game Engine, Game Engine Foundations, NC State

August 2023 - December 2023

- Built a 2D game engine in **C++** with SFML rendering, a priority-ordered event system, collision physics, and an embedded **V8 JavaScript** runtime for gameplay scripting
- Implemented client-server multiplayer over **ZeroMQ** and a thread-safe anchored **timeline** supporting pause and variable tick rates

Education

North Carolina State University

Raleigh, NC

Bachelor of Science, Computer Science — GPA: 3.98

May 2024

Technical Skills

Languages: Python, Go, Java, C++

AI Systems: LlamaIndex ReAct Agents, LLM Tool Design, MCP, A2A, AWS Bedrock

Backend & Infrastructure: FastAPI, Spring Boot, React, Kubernetes, Docker, PostgreSQL, Event-Driven Microservices

Security: STRIDE Threat Modeling, CVSS v3.1, OAuth2/OIDC, mTLS, RBAC, Vault, AI-Driven SecOps